

Jiseok Chae | 채지석

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Academic Positions

Natural Science Research Institute, KAIST

Postdoctoral Researcher

Mar. 2026 – Present

Education

KAIST

Integrated M.S./Ph.D. Program, Mathematical Sciences

Mar. 2021 – Feb. 2026

Thesis : Two Theoretical Advances in Generative Models: minimax optimization methods with improved convergence and provable benign overfitting

Advisor : Prof. Donghwan Kim

KAIST

Bachelor of Science, Mathematical Sciences

Mar. 2016 – Feb. 2021

Graduated *summa cum laude*, GPA 4.06/4.3

Minor in Computer Science

Professional Interests

- Continuous Optimization for Machine Learning
- Design and Analysis of Machine Learning Algorithms
- Generative Models and Understanding How They Work

Publications

Peer-reviewed

Understanding Schedule-Free Methods in Nonconvex Optimization: Rate Guarantees and Escaping Saddles

Jiseok Chae, Donghwan Kim

40th Annual Conference on Neural Information Processing Systems

NeurIPS 2026

Rethinking Memorization–Generalization Trade–Off in Generative Models

Jiseok Chae*, Kyuwon Kim*, Donghwan Kim

*equal contribution

40th Annual Conference on Neural Information Processing Systems

NeurIPS 2026

Optoelectronic Synaptic Transistors Based on Colloidal CdSe Nanowires for Energy-Efficient Neuromorphic Computing

Woosik Kim*, Jiseok Chae*, Taesung Park, Hanseok Seo, Hyung Jin Choi, Byung Ku Jung, Seungshin Lim, and Soong Ju Oh

*equal contribution

ACS Applied Materials & Interfaces, 18(5), 2026

ACS AMI (2026)

Stochastic Extragradient with Flip-Flop Shuffling & Anchoring: Provable Improvements

Jiseok Chae, Chulhee Yun, Donghwan Kim

38th Annual Conference on Neural Information Processing Systems

NeurIPS 2024

Two-timescale Extragradient for Finding Local Minimax Points

Jiseok Chae, Kyuwon Kim, Donghwan Kim

Twelfth International Conference on Learning Representations

ICLR 2024

HEaaN-STAT: a privacy-preserving statistical analysis toolkit for large-scale numerical, ordinal, and categorical data

Younho Lee, Jinyeong Seo, Yujin Nam, Jiseok Chae, Jung Hee Cheon

IEEE Transactions on Dependable and Secure Computing, Issue 3, 2024

TDSC 2024

Open Problem: Is There a First-Order Method that Only Converges to Local Minimax Optima?

Jiseok Chae, Kyuwon Kim, Donghwan Kim

36th Annual Conference on Learning Theory (Open Problems Track)

COLT 2023

Workshops/Preprints

Anytime Convergence via Interpolation Annealing in Schedule-Free Optimization

Jiseok Chae

OPT2026: 18th Annual Workshop on Optimization for Machine Learning

NeurIPS 2026 Workshop

Tight Convergence Analysis of Alternating Gradient Descent-Ascent with Negative Momentum via Parametric SDPs

Jiseok Chae

Preprint.

Professional Experience

CryptoLab, Inc.

Research Programmer Internship

Seoul

Jun. 2019 – Aug. 2019

- Developed HEaaN.STAT, a statistical data analysis program using the homomorphic encryption scheme HEaaN. Joint work with Korea Credit Bureau

Skills

(Programming) Languages & Software

- Python (NumPy, PyTorch, ...)
- C++
- MATLAB
- $\text{\LaTeX}$

(Natural) Languages

- Korean (Native)
- English (Fluent)
- Japanese (Intermediate)

Invited/Contributed Talks

Tight Convergence Analysis of Alternating Gradient Descent-Ascent with Negative Momentum via Parametric SDPs

2026 KSIAM Annual Meeting; Korea (Scheduled)

Understanding Schedule-free Methods in Nonconvex Optimization

2026 Korean Mathematical Society Annual Meeting; Korea (Scheduled)

Constructing Convergent Methods for Minimax Optimization Problems

KIAS AI and Natural Sciences Center Seminar; Korea, 2026

Yonsei University Computational Science and Engineering Seminar; Korea, 2026

Schedule-Free Methods are Rate-Optimal in Nonconvex Optimization

2026 KSIAM Spring Conference; Korea, 2026

Rethinking Memorization–Generalization Trade–Off in Generative Models

2025 Korean Mathematical Society Annual Meeting; Korea, 2025

Stochastic Extragradient With Flip-Flop Shuffling & Anchoring

2025 International Conference on Continuous Optimization; USA, 2025

2024 Korean Mathematical Society Spring Meeting; Korea, 2024

Two-timescale Extragradient for Finding Local Minimax Points

KAIST Stochastic Analysis and Application Research Center *CoLabor* Workshop; Korea, 2023

Selected Honors, Awards, and Grants

Jang Young Sil Fellow Program (Excellence Track)

KAIST 2026

Outstanding Paper Award

KAIST Doctoral dissertation 2026

National Academic Excellence Scholarship (Science and Engineering)

Korea Student Aid Foundation (KOSAF) 2016 – 2020

Dean's List

College of Natural Sciences, KAIST Spring 2019

School of Freshman, KAIST Spring 2016

Teaching Experience

KAIST

- Head Teaching Assistant
 - MAS110 Linear Algebra for Data Science Spring 2022 – Spring 2024
 - The founding head TA of the newly established course.
- Programming Teaching Assistant
 - MAS109 Introduction to Linear Algebra Spring 2022 – Spring 2024
 - Taught MATLAB; designed course materials and programming assignments.
- Teaching Assistant
 - MAS.10002 Calculus II Fall 2025
 - MAS473 Introduction to AI with Mathematics Fall 2024
 - MAS374 Optimization Theory Fall 2021
 - MAS101 Calculus I Spring 2021 – Fall 2021

Additional Information

- Technical Research Personnel (전문연구요원) Mar. 2024 – (Feb. 2027)
- Regular member of the Korea Cube Culture United (KCCU)